Yenka practical tasks: analogue electronics

National 5 · Electronics & control · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1a) Record the supply voltage, the current and the voltage across the resistor.

- Supply 6 V; current 60 mA (0.06 A); voltage across the resistor 6 V

1b) Calculate the expected current using I = V ÷ R, and compare it with the simulation.

- I = 6 ÷ 100 = 0.06 A: the same as the simulated value

1c) Change the supply to 3 V, 9 V and 12 V. Record the results and describe the pattern.

- 3 V: 0.03 A; 9 V: 0.09 A; 12 V: 0.12 A
- Current is proportional to voltage: doubling the voltage doubles the current

2a) Record the current and the voltage across each resistor.

- I = 13 mA; V_{220} = 2.87 V; V_{470} = 6.13 V

2b) Show that the voltages add up to the supply and that the current is the same everywhere.

- 2.87 + 6.13 = 9 V
- The ammeter reads 13 mA wherever it is placed

2c) Calculate the expected values and explain any difference.

- I = 9 ÷ 690 = 0.0130 A
- V_{220} = 2.87 V, V_{470} = 6.13 V
- Small differences come from rounding the current too early

2d) Predict what happens if the two resistors swap places, then test it.

- No change: the current and each resistor's voltage stay the same

3a) Record the branch currents and the total current.

- I_{100} = 0.12 A, I_{300} = 0.04 A, total 0.16 A

3b) Show that the total current is the sum of the branch currents, and state the voltage across each branch.

- 0.12 + 0.04 = 0.16 A
- 12 V across each branch

3c) Add a third resistor of 300 Ω in parallel. Predict the new total resistance and the effect on the total current.

- $1/R_T = 1/100 + 1/300 + 1/300 = 5/300$, so R_T = 60 Ω
- The total resistance falls, so the total current rises (to 0.2 A)

4a) Calculate the total resistance before you build the circuit.

- $R_{parallel} = (400 \times 600) \div 1000$ = 240 Ω
- R_T = 240 + 100 = 340 Ω

4b) Measure the total current and the voltage across the parallel section, and compare with calculated values.

- I = 20 ÷ 340 = 0.0588 A (58.8 mA)
- $V_{parallel}$ = 0.0588 × 240 = 14.1 V

4c) Explain why the current through the 400 Ω resistor is greater than through the 600 Ω resistor.

- Both branches have the same voltage, so the branch with the lower resistance lets more current through (I = V ÷ R)

5a) Build a 50 Ω combination in Yenka and prove it with meter readings.
- Two 100 Ω resistors in parallel
- e.g. on 9 V the current is 0.18 A: R = 9 ÷ 0.18 = 50 Ω

5b) Explain why your solution works.
- Each resistor has the full voltage across it, so twice the current flows compared with one 100 Ω resistor: the resistance is halved

5c) Extension: make 25 Ω using four 100 Ω resistors, and 150 Ω using three.
- 25 Ω: all four in parallel
- 150 Ω: two in parallel (50 Ω) in series with the third

6 Complete the results table.
- Total resistance 30 Ω
- Current 0.2 A
- Parallel voltage 3 V
- Current in one 30 Ω resistor 0.1 A

7 Higher thinking: investigate and explain what happens when (a) a parallel resistor is removed, (b) a series resistance increases. Then build a circuit that draws exactly 0.2 A from 12 V.
- (a) The total resistance rises, so the total current falls
- (b) The total resistance rises, so the current falls and the other resistors get less voltage
- 0.2 A from 12 V needs 60 Ω, e.g. a 60 Ω resistor or 100 Ω in parallel with 150 Ω